Vadim Vadanici

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Education

Imperial College London — London, UK

MEng Biomedical Engineering (Computational Bioengineering Stream) — *Expected 2028*

- Predicted First Class Honours; top of a 120+ cohort in Year 1 Mechanics & Electronics (79% overall, 83% Mechanics).
- Undergraduate Researcher, Dario Farina Lab (2025–2026): machine learning on EMG hand-gesture recordings.
- Year 3 modules: Probability, Statistics & Data Analysis; Statistical Signal Processing & Inference; Computational Continuum Mechanics.

École Active Bilingue Jeannine Manuel — Paris, France

French Baccalaureate, International Option (English) — *2024*

- Highest distinction (17.61/20) — perfect scores in Mathematics, AP Calculus BC, History, English and AP Language.
- Honourable Mention, French Mathematics Olympiad; Concours Général candidate (Philosophy, Physics-Chemistry).

Research & Professional Experience

PITOR — Paris Institute for Transplantation & Organ Regeneration — Paris, France

Research Intern — *August 2026–Present*

- Measured the accuracy and run-to-run variation of frontier LLMs diagnosing kidney-transplant rejection from Banff lesion scores: ∼30 runs over all 240 biopsies, varying temperature, reasoning mode and model version.
- Built a local LLM pipeline automating the thematic analysis of 6,000+ patient responses, with source-verified quotes.
- Preparing a first-author manuscript, continuing part-time alongside my degree.

Okeiro (AI Medtech Start-up, Chronic Kidney Disease) — Paris, France

Summer Consulting Intern — *June–July 2025*

- Developed a US market-entry strategy through competitor benchmarking and market sizing.
- Pitched the business model to venture capital investors, reporting directly to the CEO.

Trading & Quantitative Training

Optiver & Imperial College London — Trading Academy — *2026*

Month-long course on options theory, volatility and market making; implemented and optimised trading strategies in Python on real market data, competing against peers on Optiver's simulated exchange.

AmplifyME — Asset Management & Investment Banking Live Simulation — *2025*

University of Cambridge — Machine Learning & AI for Medicine Summer School (van der Schaar Lab) — *2025*

Projects

Personal Investment Research System

Python, SQLite — *August 2026–Present*

- Developed a system that researches stocks, tracks my live portfolio and grades every buy recommendation against the market after 30 days; outperforming the S&P 500 since launch.

London AI x Science Hackathon (Iterate) — Polaron & Cognition Track — London, UK

One of 150 participants selected from 1,000+ applicants — *October 2026*

- Segmented battery-electrode microstructure via multi-Otsu to predict manufacturing changes before batteries are built.
- Linked literature-based markers that differed significantly between batches to a Doyle–Fuller–Newman battery model.

Neural Manifold Learning for Brain-Imaging Biomarkers — Imperial Group Project — London, UK

Project Manager (7-member team), supervised by Prof Simon Schultz — *October 2026–Present*

- Year-long research project applying neural manifold learning to fMRI data from the UK PREVENT Dementia cohort.

Smart Elastic Resistance Bands for Physiotherapy — Imperial Group Project — London, UK

Project Manager (8-member team) — *October 2025–June 2026*

- Led the team to the Year 2 Biomedical Engineering Prize for Best Group Project: wristbands that gamify physiotherapy.
- Sole developer of the project's software: patient app, clinician dashboard, FastAPI backend, Bluetooth game.

Skills & Languages

Programming: Python (NumPy, pandas, SciPy, scikit-learn, PyTorch), SQL, MATLAB, Java, Git

Software: Excel, PowerPoint, LaTeX | **Languages:** Fluent in English, French and Romanian; German (B1)

Leadership & Extracurricular Activities

- Selected as one of 500 attendees from 5,000+ applicants for Y Combinator Startup School Paris (2026).
- Co-founder of a Quantum Mechanics club; member of the ENS Paris Ulm Maths Club.
- Former school basketball captain (10+ years); Imperial Basketball Team; EDUCAP/CAPSAA inclusive-sport organiser.
- Peer Leader and Class Delegate for international students; tutored Ukrainian students displaced by the war.
- Three exhibitions with Duchamp Prize winner Tarik Kiswanson; subject of his work *Vadim* (kadist.org/work/vadim).
- **Interests:** Poker, chess, music (DJing, piano), skiing, football, philosophy.